



# Utilities of Virtual 3D City Models Based on CityGML: Various Use Cases

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## Abstract

Virtual 3D city models are increasingly being used to model the realms of the real world for utilization in a number of applications related to environmental simulations including, urban planning, mapping the energy characteristics of buildings, noise mapping, flood modelling, etc. Apart from geometric and appearance/textural information, these applications have a requirement for complex urban semantics. Currently, a number of 3D standards are available in CAD, BIM and GIS related domains for the storage, visualization and transfer of 3D geospatial datasets. Initially, the 3D data models (such as COLLADA, VRML, X3D, etc.) were purely graphical/geometrical in nature and mainly used for visualization purposes. With the inclusion of thematic modules in OGC CityGML, the integration of geometry and semantics in a single data model paved the way for better sharing of virtual 3D city models. In spite of the availability of a wide range of 3D data standards, there are certain differences with respect to geometry, topology, semantics, LODs, etc., which complicates the integration of 3D geodata from heterogeneous sources. This paper serves to highlight the need for the innovative solutions with respect to the urban environmental related simulations primarily based on the use of virtual 3D city models. Four use cases are studied in this context namely, (1) urban solar potential estimation using CityGML models, (2) simulation of traffic noise level mapped on building walls from the urban road segments, (3) CityGML based 3D data models interoperability, and (4) 3D indoor logistics and subsurface utilities. However, for modelling majority of use cases, CityGML does not provide explicit thematic representations but provides support for extending the CityGML schema using Application Domain Extensions. In a nutshell, the study explores the semantic modelling capabilities of the CityGML for the transformation of native 3D virtual city models to one satisfying capabilities like semantic information and support towards interoperability.

**Keywords** CityGML · 3D modelling · GIS · 3D standards · Interoperability · Traffic noise

## Introduction

Recently, 3D Geoinformation has become one of the foremost research area especially in the context of crisis and disaster, environmental related simulations, facility management, urban planning to name a few. Moreover, recent advancements in 3D GIS has led to wide representation of geospatial objects in 3D, enabling storing the object characteristics such as location, shape (geometry)

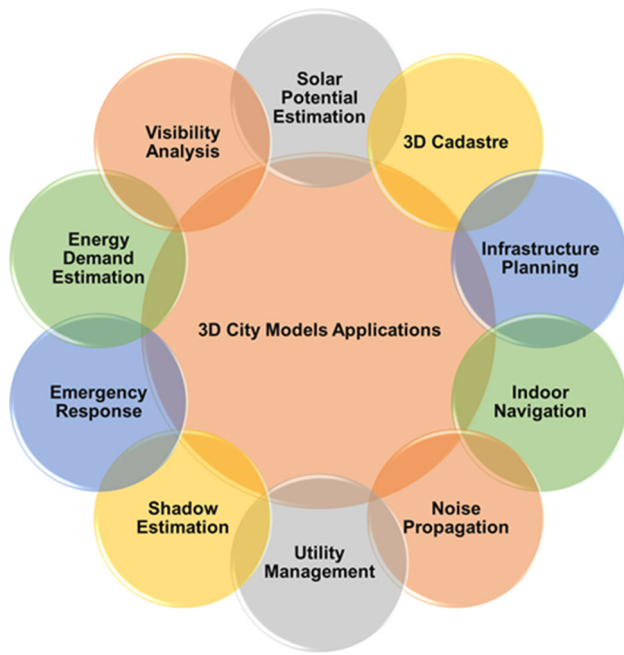
and area etc. in 3D space. Further it also encourages the exchange of such 3D datasets. The potential of GIS based system could be highly expanded through incorporation of 3D modelling of a city. Virtual 3D city models are digital representation, of spatial objects within a city, in 3D environment (Stadler and Kolbe 2007). The application domains of 3D city modelling are gradually increasing, especially in urban environment, as shown in Fig. 1 (Biljecki et al. 2015).

This leads to huge repositories of virtual representations of cities (3D models) in virtual globe software like Google Earth, NASA World Wind, etc. The common applications of virtual 3D city models exist in the areas like disaster management (Kolbe et al. 2008; Schulte and Coors 2009), planning at city level (Xu et al. 2014; Döllner et al. 2006), estimating solar potential of structures (Carrión et al. 2010;

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**Fig. 1** Application domains for 3D city models. (Source: Biljecki et al. 2015)

Krüger and Kolbe 2012; Kaden and Kolbe 2013; Saran et al. 2015), mapping of noise (Czerwinski et al. 2007), etc. Biljecki et al. (2015) has provided some of the emerging list of uses cases of 3D city models as shown in Table 1.

Given the widespread use of 3D city modelling in various areas, it also has its own challenges related to semantics, geometry, topological relationships, accuracy, texture etc. which leads to issues related to integration and exchange of 3D datasets. Various approaches and methods have been proposed by researchers to support 3D data acquisition, modelling, analysis, visualization, storage and

querying (Köninger and Bartel 1998; Lee and Zlatanova 2008; Zlatanova and Tempfli 2000).

Wate et al. (2013) has described and shown the various vector based capturing techniques for 3D data with reference to Level of Detail (LOD) for objects like buildings in a typical city environment. Furthermore, a matrix for 3D GIS modelling is suggested that serves as guidelines related to 3D spatial data creation related to various CityGML LODs. In order to provide a holistic solution, the semantics level information which involves components like above ground, indoor and underground representation during 3D modelling is desired (Zhang et al. 2011). With the advancement in web based technologies, the virtual 3D city models are also being shared between myriad applications that in turn use diverse standard data formats in order to perform analysis using various inbuilt tools. Myriad data formats for 3D representation have their own pros and cons for the said application. In order to assist sharing and usage of such 3D models among various tools and applications, a platform independent and interoperable data format for 3D representation is required. The good thing about such interoperable models is that they enable seamless sharing and accessing datasets by various applications and also facilitates conversion from the given standard to native formats used by the particular 3D GIS software or to some other format as per the needs of the user. In order to solve the interoperability issue, CityGML format serve as a unifying solution. One of the important points to consider in the 3D data representation is the amount of details to be included/depicted. This depends on the data availability and the underlying application use case. LOD concept assist in providing the amount of details required in a typical 3D representation (Benner et al. 2013). CityGML is not only an semantic model but also provides for a framework to represent spatial objects at various LOD

**Table 1** 3D city model emerging use cases. (Source: Biljecki et al. 2015)

| Use cases                                     | Description                                                                                                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Understanding synthetic aperture radar images | Methods involve simulating the acquisition and analyzing SAR scattering effects with building of different configurations                                                                      |
| Lighting simulations                          | 3D models in planning the lighting of landmarks and scenarios developed to optimize associated costs                                                                                           |
| Emergency response                            | Disaster and emergency response by providing best position for fire fighters etc                                                                                                               |
| Radio-wave propagation                        | 3D city models to predict Wi-Fi coverage                                                                                                                                                       |
| Routing                                       | Indoor and outdoor routing for indoor emergency response facilitation                                                                                                                          |
| Volumetric density studies                    | 3D models for volumetric analyses (vertical and horizontal distribution) over an urban space and use it for modeling the dispersion of urban pollutants                                        |
| Archaeology                                   | 3D model for modeling of archeological 3D objects and their attributes, analyzing development of sites over time, urban reconstruction of ancient cities and testing reconstruction hypotheses |
| Indoor–outdoor                                | Integration of Indoor–outdoor data models                                                                                                                                                      |
| ADE                                           | Development of new Application Domain Extension under CityGML                                                                                                                                  |

(Saran et al. 2015). This facilitates the display and analysis of the spatial object at various resolutions based on user requirements. CityGML further supports the integration and merging of two such dataset of the same 3D object though varying in level of details (Kolbe et al. 2005). Appearance, geometry and scene are three components representing the attributes of the 3D model required for displaying or rendering of such objects. Geometry characteristics of 3D model is represented by series of 3D points. These points can be combined in turn to form lines and surfaces. Volume based and surface based are two broad techniques for representing 3D objects (Li 1994). Objects in surface based models are characterized by basic elements of surface like facet model, shape, grid and boundary representation (B-rep). In contrast to surface based models, the volume based models are characterized by elements like 3D TIN, constructive solid geometry (CSG), array of 3 dimensions and so on. In boundary representation, the underlying surface is characterized by series of faces. Regarding vertices in the particular phase, it can range from 2 to 3 depending on whether the element is line or surface (McHenry and Bajcsy 2008). Objects in boundary model is represented through mesh to describe the object boundary (Murali and Funkhouser 1997). Further, this model also support topological characteristics. Drawback of B-rep is the utilization of B-spline function to specify the non-planer surfaces which are extremely demanding with respect to computation needs. Table 2 provides a high level view of various 3D data models created and being used in various applications. COLLADA, VRML and X3D are prominent data models useful for visualizing 3D datasets used mainly for 3D modelling and accurate rendering. Building Information Modeling (BIM) on the other hand represent building characteristics using its semantics and shape properties. It also includes the representation in digital form of the buildings surroundings. Example of 3D BIM models are gbXML (green building XML) and IFC (Industry Foundation Classes). Further these are semantics based and object-oriented models having more data about underlying buildings, its elements and relationships among them. Both these models are extensively used for exchange and storage of 3D objects so as to represent both shape and semantics. On the other hand gbXML also support information regarding simulation of energy potential for buildings.

In geospatial field, CityGML is widely used as exchange and storage format especially for the urban planning. The 3D data models mentioned in the Table 2 are from various organizations that use various standards with diverse representation for depicting the said 3D model. While Keyhole Markup Language (KML) allows only basic geometric elements like point, line, polygon and combination of geometries, VRML and X3D enables storage of free-form surfaces. COLLADA model in comparison to KML used mesh as a means to represent geometry information. For 3D representation gbXML and IFC uses closed shell geometry (Sokolov and Crosby 2011) and constructive solid geometry (CSG) and B-rep (Thein 2011) model respectively. CityGML uses B-Rep model for 3D representation of objects in GIS environment (Gröger et al. 2012). The above mentioned data models also have difference with reference to storing semantic aspect and schema design. While on one hand IFC, CityGML and gbXML are semantics enabled COLLADA and X3D are used more for rendering and data exchange of 3D models. Such issues leads to integration problem of these data models.

However, transformation of information among BIM, CAD and GIS is difficult and also leads to information loss. Table 3 provides the difference between various 3D data models. The 3D models in GIS, BIM and CAD are differentiated on following points (Stoter et al. 2011):

- XML dependence
- Geometry
- Semantics
- LOD
- Texture
- Web rendering support

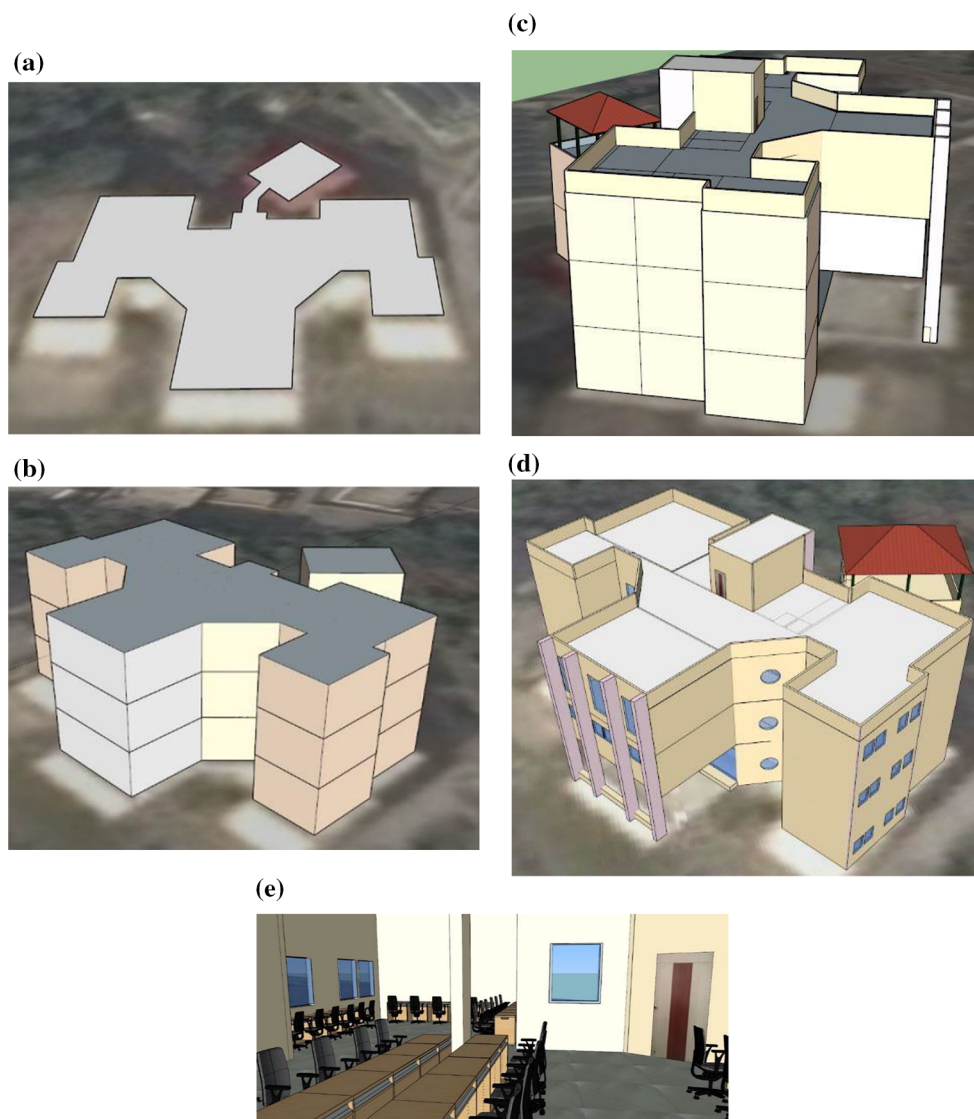
For storage, 3DCity Database is used, named '3DCityDB', which is available for two versions namely oracle spatial and PostGIS. 3DCity database is an open source product and allows storage and analysis of the CityGML based virtual 3D objects inside PostGIS DB. It supports CityGML schema including necessary support for Data import and export provided by 3DCityDB product. Main characteristics of 3DCityDB are, it is semantically rich, hierarchically structured model, supports all five level of details (Fig. 2), fully supports CityGML1.0 and 0.4.0, and

**Table 2** Potential use cases of GIS, CAD, and BIM based data models. (Reproduced with the permission from Kumar 2015)

| Domain | Data models (3D)       | Applications                                              |
|--------|------------------------|-----------------------------------------------------------|
| CAD    | OBJ, DXF, 3DS, COLLADA | Geometrical modelling and visualization                   |
| CG     | VRML, X3D, COLLADA     | Web based visualization and representation                |
| BIM    | IFC, gbXML             | AEC (Architecture, Engineering and Construction) industry |
| GIS    | GML, CityGML, KML      | Exchange and rendering of 3D spatial objects              |

**Table 3** Comparison of 3D data models. (Reproduced with the permission from Kumar 2015)

| Standard/<br>criteria | KML                   | VRML              | X3D               | COLLADA          | CITYGML              | IFC                  | gbXML                |
|-----------------------|-----------------------|-------------------|-------------------|------------------|----------------------|----------------------|----------------------|
| XML based             | XML based             | Test based        | XML based         | XML based        | XML based            | Test based           | XML based            |
| Geometry              | Primitive<br>geometry | Geometry<br>nodes | Geometry<br>nodes | Mesh             | B-rep                | CSG                  | Shell                |
| Semantics             | No semantics          | No semantics      | No semantics      | No semantics     | Rich in<br>semantics | Rich in<br>semantics | Rich in<br>semantics |
| LOD                   | Not supported         | Basic support     | Basic support     | Not<br>supported | 5 discrete LODs      | Not supported        | Supported            |
| Texture               | Basic support         | Rich textures     | Rich textures     | Rich textures    | Basic support        | Not supported        | Not supported        |
| Web rendering         | Supported             | Supported         | Supported         | Supported        | Supported            | Not Supported        | Not Supported        |

**Fig. 2** **a** Level of detail (LOD) 0, **b** level of detail (LOD) 1, **c** level of detail (LOD) 2, **d** level of detail (LOD) 3, **e** level of detail (LOD) 4

xml validation of the CityGML documents is also possible etc. (Kunde et al. 2013).

Interoperability of the 3D city models is very essential to further the scope of use and its future uses. Interoperability basically refers to the ability for system or its modules to access the distributed heterogeneous datasets in various format through a single interface (Schell et al. 2000).

### Pros and Cons of 3D Data Models

Pros and cons of using the 3D data models are:

1. Normally 3D city model delivers a true picture and real scene of the ground and enable the planners to view the locations of services and real places in an intuitive and use-friendly way.
2. 3D data models provide and reuse multi-scale data in different applications i.e. there is data consistency. This is supported by multi-representation data models e.g. LOD concept of CityGML.
3. With 3D model, it is possible to view the object from different perspectives.
4. 3D geometry of the features is useful to model shadows on the features.
5. The appearance and textural information provides for more realistic visualization.

However, there are also some negative consequences associated with the use of 3D data models. Such as high cost of production, the inaccurate models according to the real surroundings, enormous amount of time for manual work for data creation, and problems of data validation.

Ross (2010) described classification of 3D application use cases of three types i.e. depending on geometry, including both geometry and semantics details and depending on domain specific extension including external datasets like in case of estimating noise emissions. Based on the same taxonomy four use cases using CityGML are illustrated in this paper namely, (1) urban solar potential estimation utilizing CityGML models, (2) simulation of traffic noise level mapped on building walls from the urban road segments, (3) CityGML based 3D data models interoperability and (4) 3D indoor logistics and subsurface utilities.

### CityGML at Semantic Level for Urban Energy Conservation Strategies

In this study IKONOS satellite image of Indian Institute of Remote Sensing (IIRS), Dehradun, India was used after performing preprocessing operations like rectification and

georeferencing with respect to field based ground control points.

From the satellite image the footprints of the buildings were extracted. Further the building of Geoinformatics Department in the campus was used for carrying out the energy conservation strategy investigation. Figure 3 shows the methodology flowchart. The model outputs are presented in Fig. 4a, b.

Thus, the obtained virtual model in native format has been transformed into open data format which has been further integrated with energy application specific information. The UML diagram of CityGML energy ADE for urban solar potential indicators is given in Fig. 5.

This facilitates rigorous and insightful queries which indeed exploits the semantic and geometry capabilities open data standard such as CityGML and can't be done using 3D model without semantics details. Some of the examples of semantic queries are (1) finding out the number of roofs and walls that are getting solar radiation for maximum interval during the whole year and (2) what is the effect of seasons changes i.e. winter and summer seasons on the walls and roofs getting the solar radiation for maximum interval during the whole year. Such queries provide valuable information about solar potential assessment and hence will be useful for utilizing the solar energy

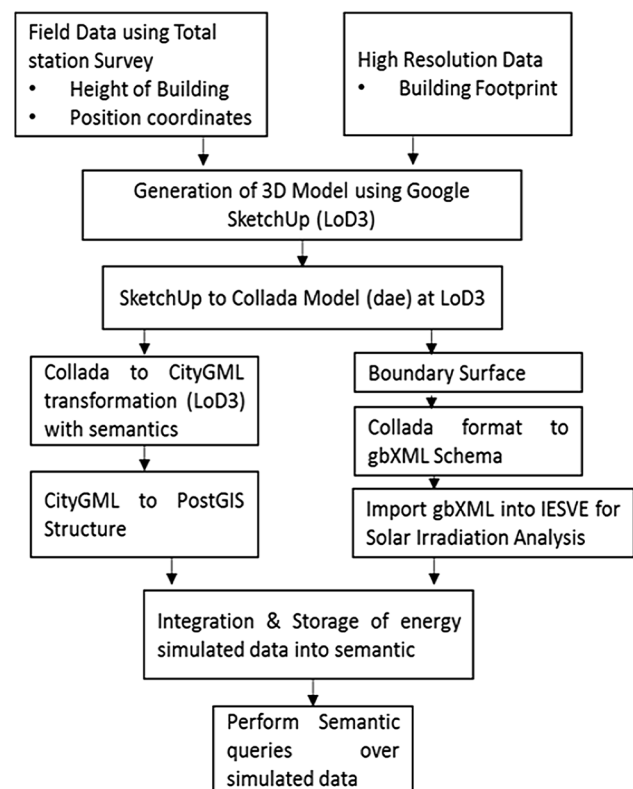
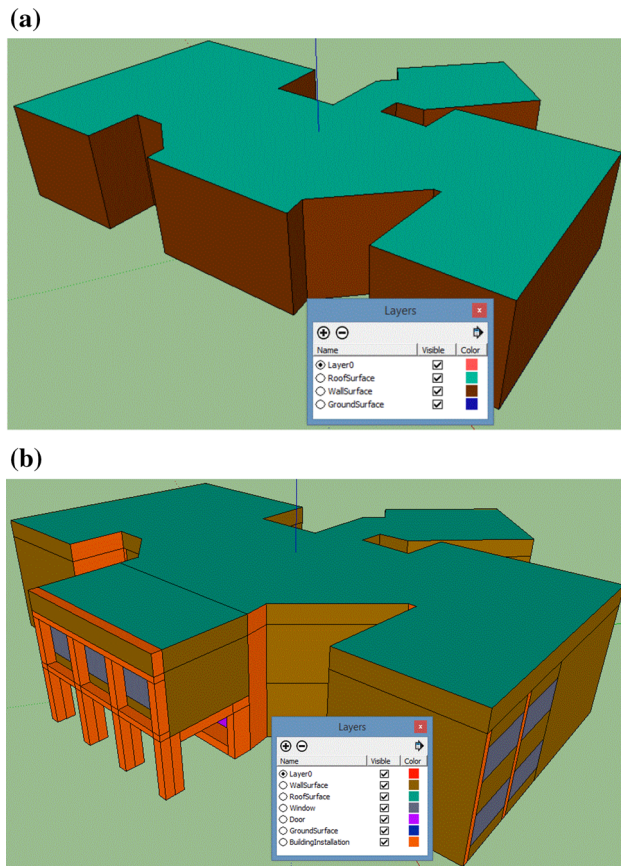


Fig. 3 Methodology flowchart



**Fig. 4** **a** Building model at LOD2, **b** building model at LOD3 with semantics and geometry

by mounting solar panels on the resultant walls and roofs and also leads to saving electricity used for lighting.

## Traffic Noise Modelling (TNM) Using CityGML

Noise has always been a main cause of irritation and distress for the people in metro cities. The main sources of noise on a vehicle include (1) engine noise (2) noise from transmission and silencers (3) noise due to road surface contact etc. The important factors that reduce the overall noise level include (1) wind as a barrier (2) road surface absorption (3) noise barrier walls etc. (Lamure 1986). A WebGIS framework was implemented for simulation of traffic noise level mapped on building walls using the data from PostGIS. Finally spatio-temporal semantic analysis to quantify the effects in terms of threshold noise level, total roofs and wall affected from the beginning to the day end was carried out. A part of the Dehradun City, Uttarakhand, India was chosen for rebuilding of 3D virtual model including display and investigation. Calixto et al. (2003) has defined three criteria's to adequately represent the model as mathematical expression so as to

ensure satisfactory estimation of noise levels in statistical terms. They are (1) must be easy to use by planners in urban domain. (2) must take for the calculation of noise levels, the datasets that are simple to obtain and (3) the model must have provision for including the acceptable outcomes based on user defined requirements. Traffic Noise Modeling (TNM) estimates the amount of noise caused by traffic coming out from the source been quantified at the emitted with flow of traffic having range of velocities. For the Indian case, deliberation of possible types of vehicles is mandatory for creating a real model for estimating traffic noise. Keeping this in mind the existing Calixto Model (Calixto et al. 2003) has been modified and presented as mentioned under:

$$Leq(A) = a * 10 * \log(Qeq) + k \quad (1)$$

$$Qeq = Q(1 + \sum(Ni + Vpi)/100) \quad (2)$$

Here

- Q denotes the total vehicle's count.
- Ni denotes the multiplication factor so as to estimate the actual traffic movement keeping in mind the role of heavy vehicles in the noise levels.
- Qeq denotes corresponding total amount of vehicles,
- a and k represents the regression coefficients and
- Vpi denotes the number of type of vehicles in percentage.

A total of 23 sample points were collected for the TNM. Moreover, they were collected in 2 of the hectic places in the Dehradun city. The locations where the sample points were collected from the above mentioned sites are depicted in Fig. 6a, b respectively. Additionally, an audio was recorded at each sample point and processed using Virtual Sound Level Meter (VLSM) v0.41 to estimate  $Leq(A)$  i.e. the A weighted noise level. The VLSM window along with an output generated after processing one of the sample audio is shown in Fig. 7.

## Web Architecture

For the display and semantic analysis of the 3D model on the Cesium globe based viewer a fat client-server based web architecture has been used. Cesium library supports the option of rendering and layering of high resolution imagery as web services from various available known services (INC 2015). The high level architecture adopted for WebGIS framework is depicted in Fig. 8. Here the web service was implemented using JSON format using the AJAX call (XmlHttpRequest protocol)

## Web Server

It allows:

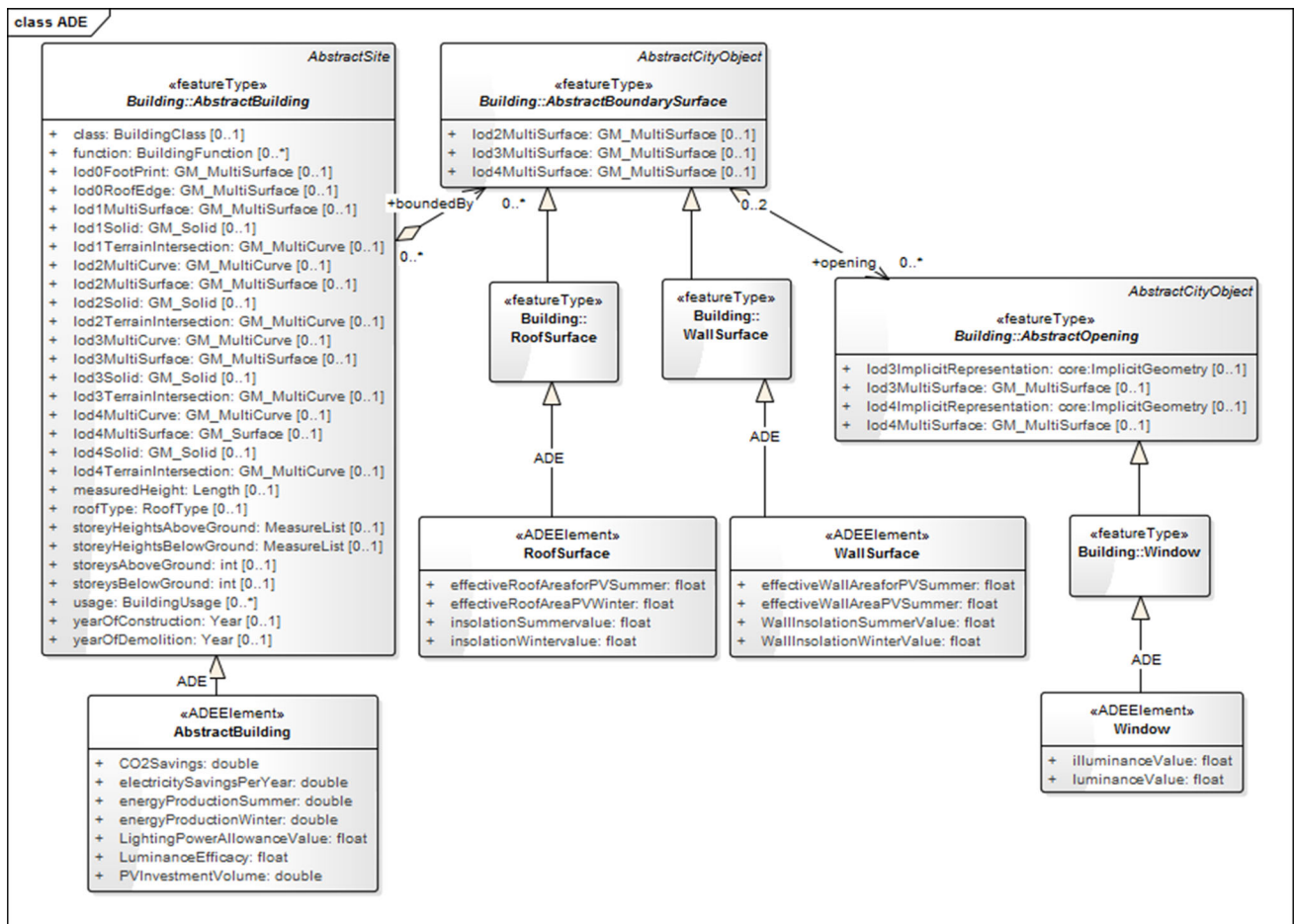
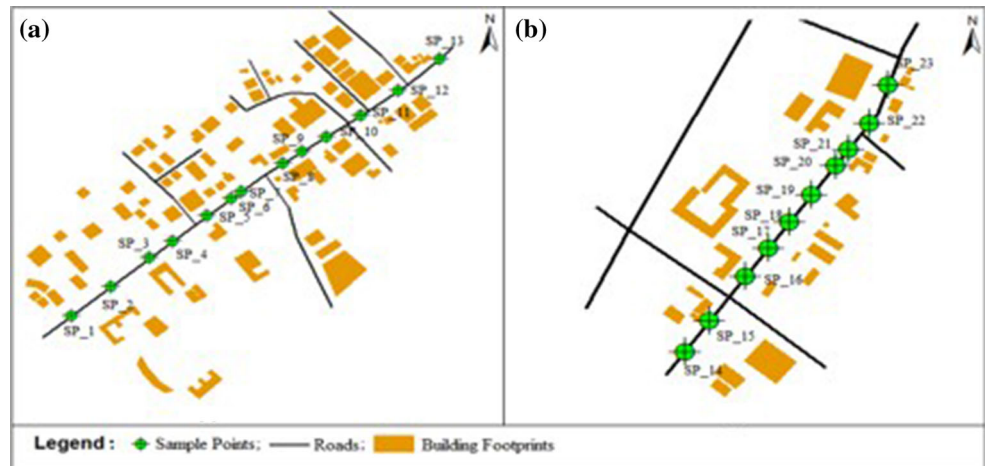


Fig. 5 UML diagram of CityGML energy ADE for urban solar potential indicators

Fig. 6 Sample points for TNM  
a site 1 (Rajpur Road), b site 2 (Saharanpur Road)



1. Spatial Database Interaction (PostGIS + 3DCityDB)
2. Transformation of the 3D vector feature and other attributes as JSON based response.

Spatial queries were used for the retrieval of building features in 3D format from the underlying City Database in 3D format making use of the object location as a predicate

in query. The result of query include parameters like name of building including construction year and level of TNM, apart from the geometry from the underlying spatial database on user request. The response is in the JSON format which is transmitted to the client browser for display and analysis at his end (Fig. 8).

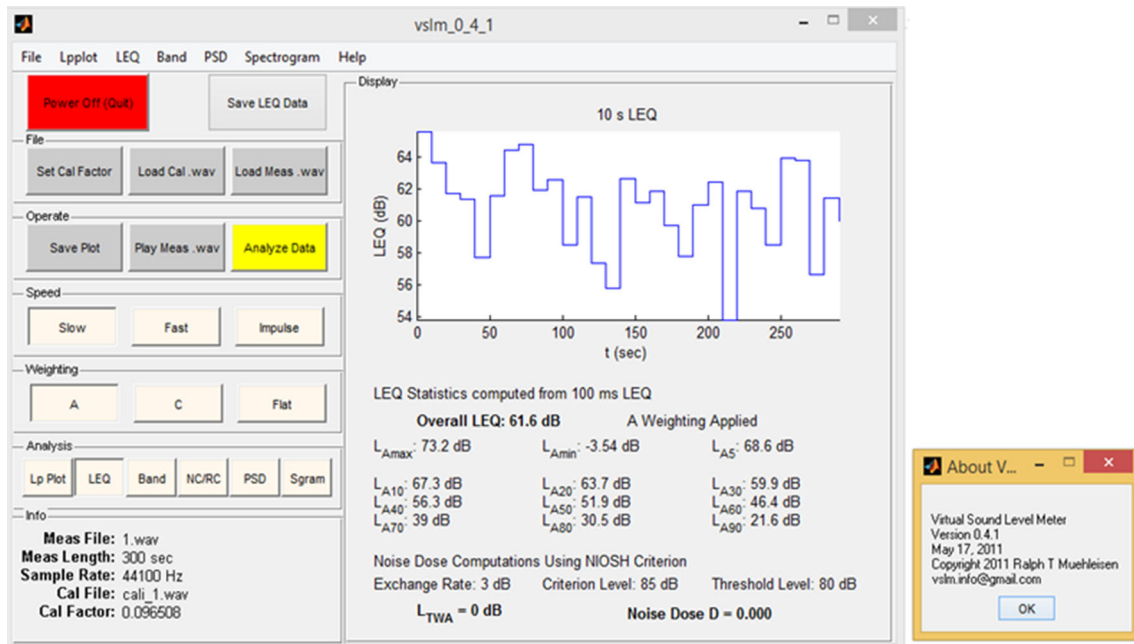
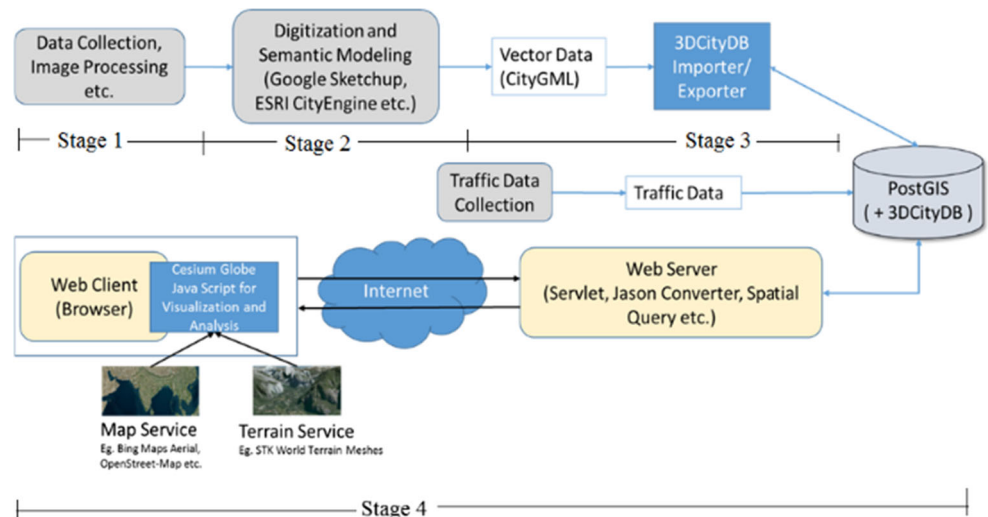


Fig. 7 VLSM v0.41 with an output window for one of the sample audio

Fig. 8 Proposed architecture traffic noise mapping and visualization



## Cesium Web Client

The customized web based client designed and developed using cesium provides visualization functionality in two forms which are 3D visualization based on Voxels and 3D Visualization at Semantic level supporting levels of details like LOD0, LOD1 and LOD2. The developed web client provides visualization and analysis capabilities for the TNM.

## Distance Attenuation

One of the main factors contributing towards reduction in noise levels is determined by the distance of the point of observation. To calculate the distance in 3D from the point of observation to the road (represented as polyline), development of algorithm was carried out. To estimate the distance from the point of observation to line segment (road), algorithm takes into consideration following 2 scenarios:

1: The perpendicular from the point of observation to the lines i.e. the point of intersection is within the endpoints

**Table 4** Expected coefficient for vehicle types

| Type of vehicle | Coefficient |
|-----------------|-------------|
| Two wheelers    | 3.1         |
| Three wheelers  | 0.7         |
| Four wheelers   | 1.1         |
| Heavy vehicles  | 6.1         |

of the road segment. Here, the distance from the intersection point and the point of observation is the least distance.

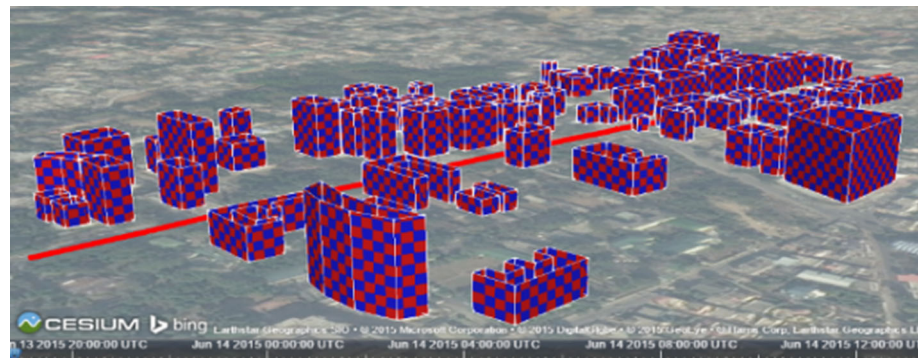
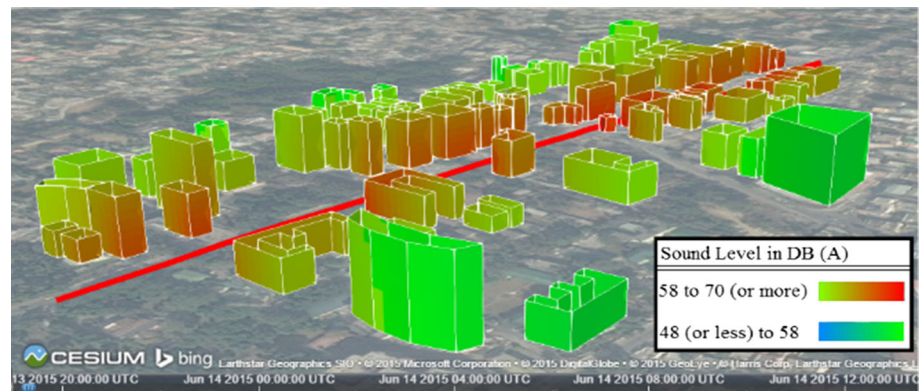
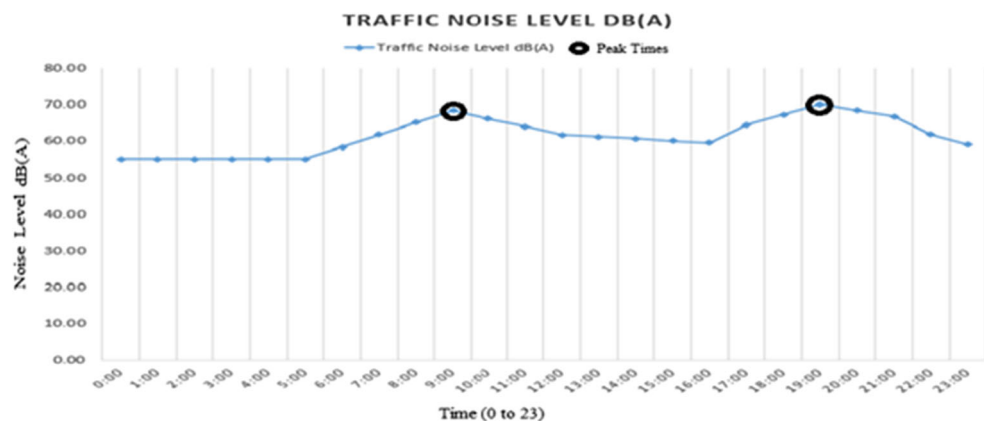
2: The perpendicular from the point of observation to the lines i.e. the point of intersection is outside the

endpoints of the road segment. Here, the least of the distances is taken of the various distances from the point of observation and endpoints of the road segment.

## Results and Discussion

### Traffic Noise Model

Table 4 depicts the expected coefficients computed following the sensitivity analysis of the TNM.

**Fig. 9** Voxel based model displayed on cesium globe**Fig. 10** Traffic noise 3D map (7 P.M.) based on voxel model**Fig. 11** Traffic noise temporal difference at 10 m distance from center of the road

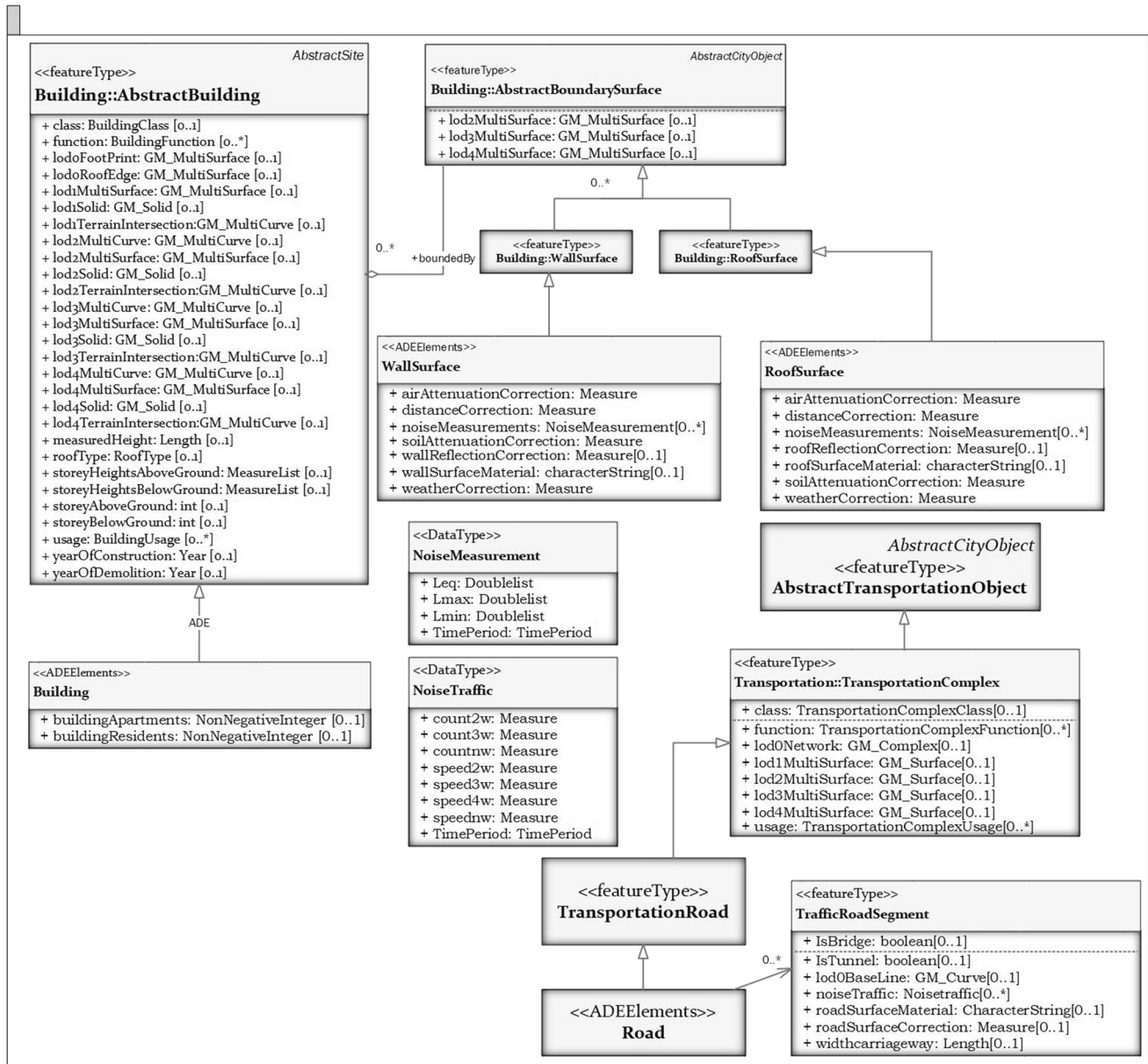


Fig. 12 UML class diagram for capturing traffic noise details

In the study area for TNM, following regression model is created and used:

$$Laeq = 37.386 + 9.839 * \log(Q * (1 + (3.5 * P_{tw} + 0.7 * P_{thw} + 1.8 * P_{fw} + 6.5 * P_{hv})/Q)) / \log(10) \quad (3)$$

Here

- Laeq represent the corresponding level of noise
- Q denotes the vehicles count
- $P_{tw}$  denotes 2 wheeler vehicles percentage
- $P_{thw}$  denotes 3 wheeler vehicles percentage
- $P_{fw}$  denotes 4 wheeler vehicles percentage

- $P_{hv}$  denotes the heavy vehicles percentage.

The models keeps in mind the myriad kinds of vehicles especially in the case of Indian roads. Mean residual error among the estimated noise level computed through the developed model and the true value was 0.08 db. While 0.72 is the correlation among the true and estimated value. RMSE (Root Mean Square Error) was assessed as 0.37 db.

### CityGML and Voxel Model Visualization

The visualization on the Cesium interface retrieved JSON objects at the client level on LOD1 and LOD2 respectively. Creation of observation points in 3D context at predefined

interval is essential for creation of 3D traffic noise maps ensuring that building walls clearly represents the traffic noise level vertical profile. To accomplish this endeavor, voxel based model was developed utilizing the LOD1 model of the study area. The same is displayed in the cesium web globe interface (Fig. 9)

### Voxels Based Spatio-Temporal Traffic Noise Mapping

In the case of voxel approach, observation point is assumed to be the center of individual grid cell that is on surface of the wall. 3D Euclidian distance from the road segment utilizing the algorithm developed from the observation point is computed. The computed distance serve as the 'noise reduction factor'. Figure 10 shows the 3D traffic noise map estimated using the above mentioned approach. Web Globe based on cesium library allows temporal display capabilities for rendering geospatial datasets. This capability was utilized for creating animation for depicting changes in the noise level due to traffic on a temporal scale (Fig. 10).

Figure 11 shows that level of noise is highest at 9 AM and 7 PM during evening. The highest level of noise is roughly 75 db (A), which is detected at 10 m distance from the center of the road. Traffic noise map producing through voxel model is a true 3D model and supports noise levels mapping at the points of observation at fixed intervals. The UML class diagram is shown in Fig. 12.

### 3D Modelling of Outdoor, Indoor and Subsurface Utilities Using Smart City Concepts

This work deals with the development of use cases and CityGML Application Domain Extension (ADE) based on indoor positioning using Wi-Fi sensor for localization, navigation through the outdoor/indoor paths on 3D models of buildings. It also looks into creation of a use case, remote identification and 3D modeling of the subsurface utility networks in the campus of Indian Institute of Remote Sensing, Dehradun, India. The datasets used in the study are the 3D COLLADA models of buildings. Following tools and instrumentation used are:

1. *Hardware and Field Instrumentation:* Ground Penetrating Radar (GPR) to detect subsurface utilities data (sewage and water pipelines) and Laser Distance Meter to measure distance (length, width, and height) of indoor spaces like doors, windows, floor height, etc.
2. *Software Tools:* The software tools used are Google SketchUp to design 3D models, PostGIS database for

3D storage of data, Enterprise Architect for Use Case modelling and ADE creation, FME quick translator to translate the models, FZKViewer for visualizing 3D CityGML models, Google Earth to get the footprints of buildings, 3DCityEditor for adding CityGML attributes in 3D model and IDS GRED HD for the processing of GPR images.

3D models cover the height field (the third dimension) over the regularly used 2D coordinates and provide better

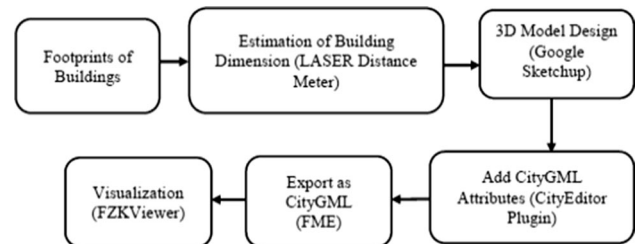


Fig. 13 Building model (3D) flowchart

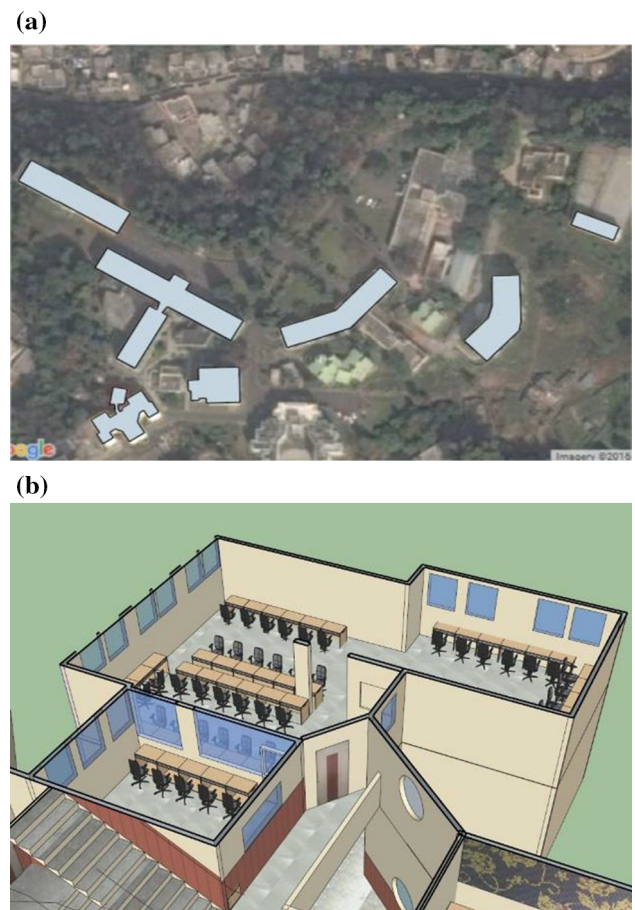
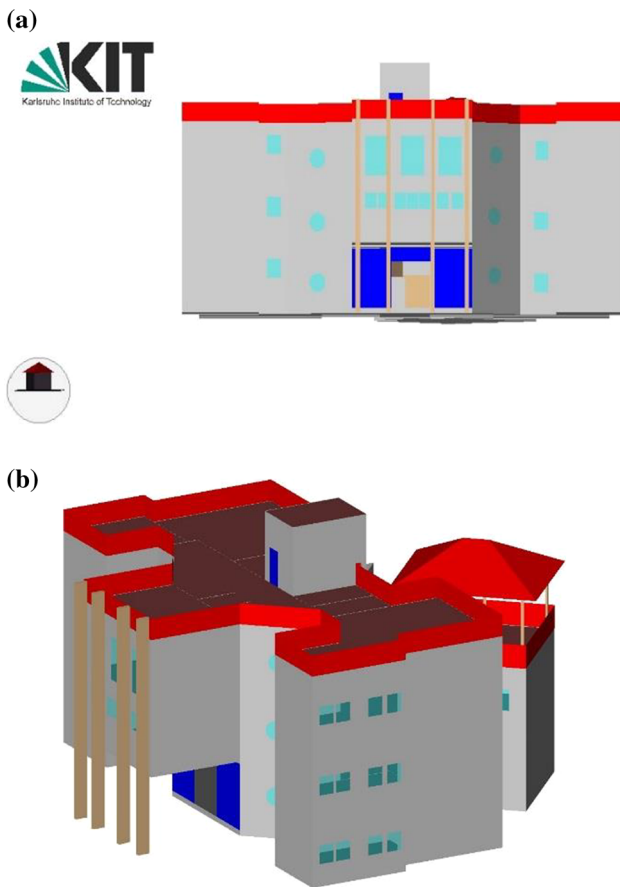


Fig. 14 a Footprints of the buildings, b LOD4 cross-sectional view

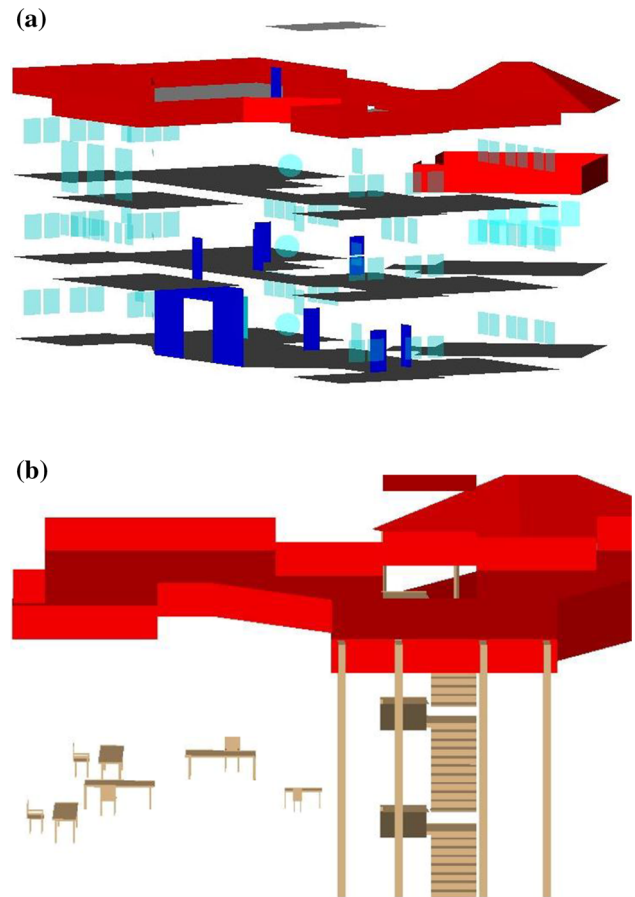


**Fig. 15** CityGML building model **a** front view, **b** perspective view

information about the real world. Figure 13 shows the process followed in the 3D building model. A building footprint and its dimensions are required to create its 3D model. Figure 14(a) shows the buildings footprints used in this study created over the Google Earth Imagery. Building footprints/dimensions were determined using a LASER Distance Meter. The subsurface utilities were visualized in ESRI ArcScene software.

Figure 14b shows the wall Surface, door, window and some of the Building Installations inside one of the buildings. Necessary attributes were added to the 3D model according to CityGML OGC standards, and the model is converted to CityGML data which can be visualized in a software package called FZKViewer. Figure 15a, b displays the CityGML building model of one of the buildings from front and perspective viewpoints respectively. This model comes under LOD4 that contains interior details such as building furniture, floor-wise surface information, stairs, interior doors and windows which are shown in Fig. 16a, b.

The UML Class diagram of the proposed Indoor Routing and positioning ADE is shown in Fig. 17.

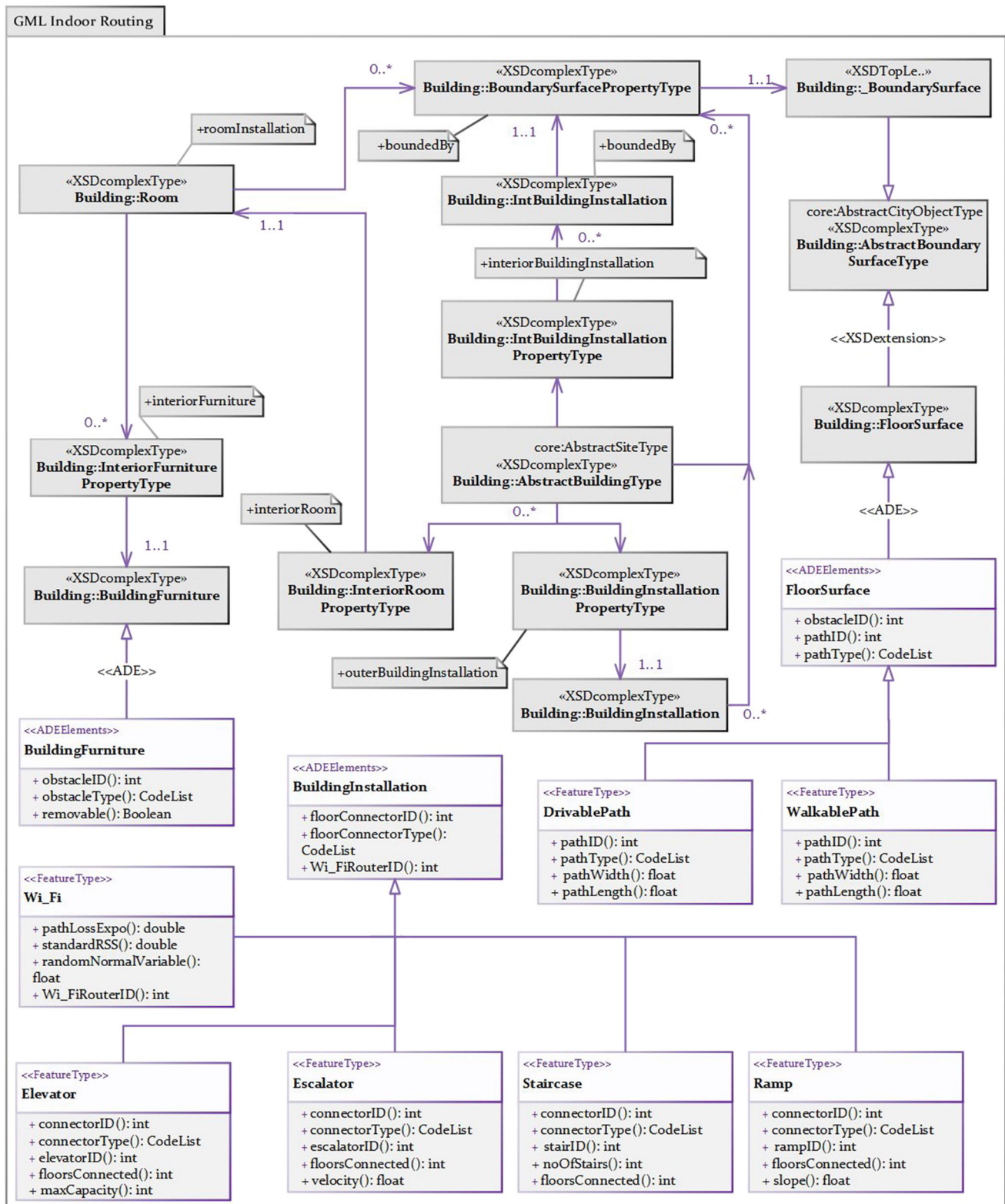


**Fig. 16** CityGML model in the fourth level of detail **a** floor Surface with interior doors and windows **b** building furniture and staircase

### CityGML Based Interoperability for the Transformation of 3D Data Models

This work deals with addressing interoperability issues among 3D data models and also showcase the mapping to transform among the same using CityGML with diverse LODs. Unified Modelling Language (UML) which is based on Object Oriented Analysis was used for representing the model at conceptual level including the mapping part. This work is carried out using Open Source Software's and Open Standards. It helps in improving the exchange mechanism of such 3D datasets with least loss of information. The study area chosen was IIRS campus, Dehradun, India. Firstly the 3D model of the study area was developed and then the 3D model was converted to COLLADA model (version 1.4.1) with different depictions. A GUI viewer for visualization of COLLADA dataset was also developed (Fig. 18).

The CityGML model created has geometry type as gml:MultiSurface representing complex surface. These surfaces increases depending on desired LOD. Based on input used and the output model generated from

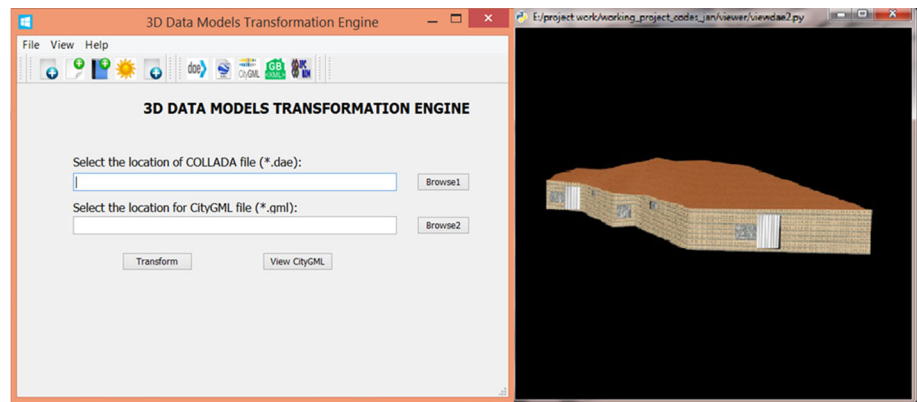


**Fig. 17** UML class diagram of the proposed indoor routing and positioning ADE

conversion, it is observed that number of basic elements of geometry representing surface in COLLADA model is similar to that of CityGML model. In case of LOD0 i.e.

representing footprint of buildings, which is depicted as geometry having 4 vertex is similar to the surface generated in CityGML represented through 4 points. In case of

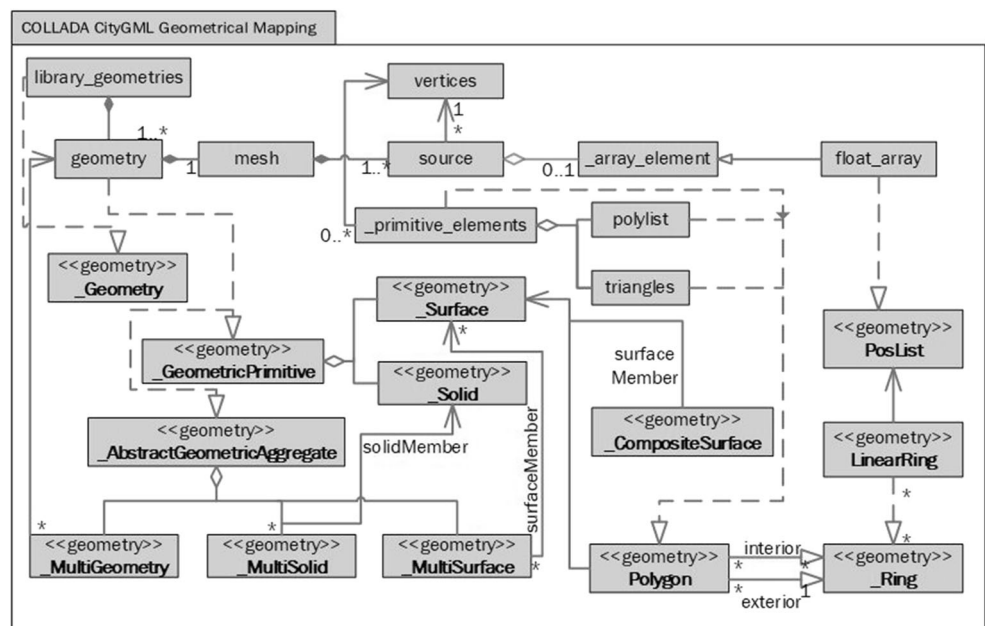
**Fig. 18** Screenshot of software and COLLADA viewer (in LoD3). (Reproduced with the permission from Kumar 2015)



**Table 5** Campus Buildings' related information at CityGML LOD2

| # | Campus buildings                  | Number of bldg: GroundSurface | Number of bldg: WallSurface | Number of bldg: RoofSurface |
|---|-----------------------------------|-------------------------------|-----------------------------|-----------------------------|
| 1 | Main building (IIRS)              | 1                             | 293                         | 1                           |
| 2 | Geoinformatics Department. (IIRS) | 1                             | 76                          | 2                           |

**Fig. 19** UML diagram for geometrical mapping of COLLADA and CityGML (up to LOD-3)



LOD1 for CityGML, the model of the IIRS campus building was depicted through 6 surfaces which is analogous to COLLADA model having 6 basic elements also. Further in case of CityGML LOD2 and LOD3 consisting of 29 and 169 surfaces for LOD2 and LOD3 respectively is similar to the one represented in COLLADA model which is being used as input. On converting the COLLADA model to CityGML, increase in the file size (about 3–4 times or more than the original size) is observed. COLLADA model in LOD2 and LOD3 are of size 16 KB and

43 KB respectively. The corresponding model on conversion to CityGML has increased file size to 37 KB and 194 KB for LOD2 and LOD3 model in CityGML respectively. On comparison to conversion by FME software to CityGML, it is observed that difference in size of file generated is 7 KB and 12 KB for LOD2 and LOD3 respectively. In order to reduce the file size for CityGML output after transformation of geometry, there is a huge need for optimization of the same. The time required to conversion by model developed is 0.075 and 0.289 s for

LOD2 and LOD3 model respectively. It is similar to one generated from FME software.

The IIRS campus model in 3D at LOD2 and LOD3 was used for conversion from CityGML to COLLADA. Table 5 highlights the semantic and geometric related information of the buildings.

Figure 19 shows the UML diagram for Geometrical Mapping of COLLADA and CityGML (up to LOD-3).

## Conclusions

3D city models are currently being used in dozens of application domains for diverse purposes. This diversity in the increasing number of applications makes it difficult to keep track of the utilization of 3D city models in different domains. In this overview of applications of virtual 3D city models, we have attempted to highlight the specific areas of research depicting how the same form of geo-information can be possibly used in different ways. The focus of the study was on the practical application of multi-LOD datasets in the fields of simulation namely, energy and noise simulation. The study was clearly divided into two crucial aspects: interoperability and application that are commonly associated with the use of 3D geoinformation. The future of 3D city models is determined by their ability to adjust the 3D data models behind them with a higher degree of Interoperability. The case of interoperability mapping for CityGML and COLLADA is beneficial for deriving semantics from the COLLADA geometry and integrating it in CityGML which can be later utilized in spatio-semantic analyses.

The presented study effectively encourages to address further future research questions which are increasingly sought by different stakeholders involved in integrated planning and decision making. From planner's side, he or she can make suggestions to the citizens by highlighting the most livable part of the city while at the same time posing towards them the energy conservation of that city quarter. While on the household side, he or she can clearly observe the trade-offs between the amount of wealth to be invested for healthier and sustainable living conditions.

Therefore, on one hand, addressing interoperability between different data standards would encourage more flexible use of native data formats while on the other hand the knowledge gained through application studies would guide urban planners to make decisions.

Seeing an increase in the demand of 3D city models, we expect new applications to emerge in the future. Recent advancements in BIM, GIS and graphics domain appear to play a prominent role in the widespread adoption of 3D city models in different simulations. However, it is interesting to note that the majority of 3D applications focus on

buildings, and not many applications makes use of other thematic classes, such as vegetation and bridges. We expect that, in the future, more applications will take benefit of other thematic features as well. With CityGML supporting LOD4 representation, we anticipate an increase in the use of indoor 3D models for indoor navigation and routing.

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